

AGRIAI 2.0

INTELLIGENT FARMING ASSISTANT FOR GHANA

Volume III. API Contracts and Reliability

How every route is shaped so the product stays up

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Repository: ThaddeusHackz/agriai-v2
Document date: 18 August 2026
Classification: Internal forensic architecture pack

This document explains how AgriAI was built, which source files make the AI work, and how the public and admin APIs stay reliable when a provider is missing. Every claim below is taken from the live codebase, not from marketing copy.

1. Design rule

AgriAI APIs are Next.js Route Handlers. They accept JSON or FormData, validate early, never throw an uncaught error to the farmer, and degrade to a useful payload when a vendor is offline. This volume is the contract sheet.

2. Public AI and media routes

Route	Role
POST /api/chat	SSE conversation. Body: message, language, mode, webSearch, history, sessionId. Events: delta, done.
POST /api/vision	Crop photo to JSON disease report via detectCropDisease.
POST /api/generate	Farm scene to image data URL. Cloudflare Flux then Gemini image.
POST /api/transcribe	Audio file to text via Gemini multimodal.
POST /api/tts	Text to MPEG via ElevenLabs.

POST /api/search	Standalone web search used by UI helpers.
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3. Chat request contract in detail

- message required, trimmed, max 2000 characters. Empty returns 400.
- language must match en|tw|ga|ee|ha|fr or it becomes en.
- mode must be expert or agent or it becomes standard.
- webSearch boolean. If true, searchWeb runs before the model.
- sessionId clipped to 64 chars, or anon_ plus a time token.
- history filtered to user/assistant strings, last 10 kept.
- Invalid JSON returns 400 Invalid JSON body.
- Success is not JSON. It is text/event-stream with no-cache, keep-alive, X-Accel-Buffering no (so nginx does not hide tokens).
- done payload: text, demo, sources, searchDemo, provider.

4. Why chat stays "perfect" under failure

Perfect here means the farmer always gets a structured stream and a persisted turn. Gemini stream errors are caught, logged as [chat] Gemini stream failed, trying Cloudflare. Cloudflare errors are caught the same way. Local answer sets demo true and clears sources so the UI does not fake citations. Persistence happens after generation so even offline answers are in the admin transcript. request.signal aborts upstream work if the farmer hits stop.

5. Data and utility routes

Route	Role
GET /api/weather	5 day forecast. OpenWeather then Open-Meteo then curated city data.
GET/POST /api/prices	Market board read and refresh with optional live intel.
GET /api/config	Public site settings the home page needs (no secrets).
GET /api/health	Liveness for Render.
POST /api/feedback	Thumbs on an answer.
GET /api/history	Session transcript for the returning browser.
POST /api/subscribe	Newsletter into subscribers[].
POST /api/contact	Contact form into contacts[].
POST /api/track	Visit analytics.
GET /api/images	Unsplash crop photos when a key exists.

6. Admin API surface

All of these sit under `app/api/admin/`. They share cookie auth from `lib/auth.ts`.

- login, logout, me: session life cycle. Login is rate limited.
- settings: read and write site copy, colors, section flags, AI model, temperature, maxTokens, three prompts.
- analytics: chats, messages, visits, top questions, feedback counts.
- users: create editors and admins, reset passwords.
- messages: list and delete conversations.
- knowledge: CRUD for offline Q and A used by localAnswer.
- prices: CRUD for the public board.
- subscribers and feedback: lists and CSV minded exports.
- reset: danger zone wipe and reseed.

7. Weather reliability pattern

Weather copies the AI ladder. OpenWeather is primary when `OPENWEATHER_API_KEY` is set. Open-Meteo is a no key scientific fallback. Seeded city advice is last. Cities in the product are Accra, Kumasi, Tamale, Takoradi, and Cape Coast. The UI never shows a hard crash page because the route always has a payload.

8. Environment contract (what must be set for live AI)

- `GEMINI_API_KEY` (or Google aliases): live chat, vision, transcription.
- `CLOUDFLARE_API_KEY` + `CLOUDFLARE_ACCOUNT_ID`: Llama fallback and Flux studio.
- `OPENWEATHER_API_KEY`: best weather.
- `TAVILY_API_KEY`: best search citations.
- `ELEVENLABS_API_KEY`: spoken answers.
- `UNSPLASH_ACCESS_KEY`: crop photos on the price board.
- `DATABASE_URL`: survive Render disk resets.
- `ADMIN_EMAIL`, `ADMIN_PASSWORD`, `ADMIN_NAME`: first admin seed.
- `NEXT_PUBLIC_SITE_URL`: canonical links, sitemap, robots.

Missing keys do not crash boot. Each helper returns null or a demo structure. That is the operational reason the site can be demoed in a classroom without secrets.

9. Validation habits that keep APIs clean

Routes parse JSON in try/catch. Strings are typeof checked. Arrays are filtered. Language and mode are whitelisted, never trusted from the client as free text in system position except after they pass the regex. Image generate rejects off topic prompts with 400. Vision rejects tiny or huge payloads. TTS clips to 1500 characters. These boring checks are why the APIs look stable in production.

10. Forensic conclusion of Volume III

The APIs work well because Thaddeus Tagoe treated every vendor as optional and every farmer request as something that must end in a typed, persisted, explainable result. Streaming, fallbacks, caps, and admin CRUD are one system, not plugins glued on later.